

COURSE TITLE : NATURAL RUBBER PRODUCTION
COURSE CODE : 3091
COURSE CATEGORY : B
PERIODS/WEEK : 5
PERIODS/SEMESTER: 75/3
CREDITS : 5

TIME SCHEDULE

Module	Topics	Period
1	History, sources, and importance of Natural rubber	17
2	Preservation and concentration of NR latex	19
3	Processing of NR latex in to dry marketable forms .	18
4	Specialty forms of Natural rubber and reclaim rubber.	21
TOTAL		75

COURSE OUTCOME

Sl.	G.O	Student will be able to
1	1	To understand the different rubber yielding plants , their propagation methods , extraction of latex and statistics and study of NR production .
	2	Comprehend the composition preservation and concentration of NR latex
2	1	To comprehend the processing of NR latex in to dry marketable forms
3	1	Understand the production, properties and applications of specialty rubbers and reclaimed rubbers .

SPECIFIC OUTCOME

MODULE I HISTORY, SOURCES, AND IMPORTANCE OF NATURAL RUBBER

- 1.1.0. To understand the different rubber yielding plants , their propagation methods , extraction of latex and statistics and study of NR production .
- 1.1.1. State the importance, uniqueness of NR, major sources of NR and statistics of production and consumption in India and world.
- 1.1.2. Describe the propagation methods and different clones of rubber plants.
- 1.1.3. Define tapping and standard of capability, tapping task, tapping rest.
- 1.1.4. List the different tapping systems in use.
- 1.1.5. Describe the different types of tapping like ladder, slaughter and puncture tapping.
- 1.1.6. Explain the nature, action and method of application of yield stimulants.
- 1.1.7. Explain the method and importance of rain guarding.

MODULE II PRESERVATION AND CONCENTRATION OF NR LATEX

- 2.1.0. To understand the composition and colloidal nature of Natural rubber latex
- 2.1.1. Explain the composition and colloidal nature of NR latex.
- 2.1.2. Explain the role of microorganisms in latex and destabilization of latex
- 2.1.3. Discuss the various short term and long term preservative systems used in NR latex
- 2.1.4. List the different concentration methods of NR latex
- 2.1.5. Explain the principle, process and applications of creaming process.
- 2.1.6. Explain the principle, process and applications of centrifuging process.
- 2.1.7. Draw the diagram of Centrifuging machine and plant lay out centrifuging plant.
- 2.1.8. Compare creamed and centrifuged latex with respect to TSC, DRC and Stability.
- 2.1.9. State the BIS specification of centrifuged latex.
- 2.1.10. Explain double centrifuged latex, skim latex and skim rubber and uses.

MODULE III PROCESSING OF NR LATEX IN TO DRY MARKETABLE FORMS

- 3.1.0. To comprehend the processing of NR latex in to dry marketable forms
- 3.1.1. List the different marketable forms of dry natural rubber.
- 3.1.2. Describe the machinery and process of manufacture of Ribbed smoked sheets and Air dried sheets.
- 3.1.3. Explain the different grades of sheet rubber.
- 3.1.4. Describe the machinery and process of manufacture of Pale latex crepe and Estate brown crepe.
- 3.1.5. Explain the different grades of crepe rubber
- 3.1.6. Explain the machinery and process of manufacture of technically specified block rubber (ISNR).
- 3.1.7. State the BIS specification of ISNR grades.
- 3.1.8. State the advantages of ISNR and its current trends in Rubber Industries.

MODULE IV SPECIALITY FORMS OF NATURAL RUBBER AND RECLAIM RUBBER

- 4.1.0. Understand the production , properties and applications of specialty rubbers and reclaimed rubbers
- 4.1.1. List the different specialty forms of Natural Rubber.
- 4.1.2. Explain the production of different forms superior processing rubbers-SP RSS, SP ADS, SP, Crepe, PA 80 and PA 57.
- 4.1.3. Describe the processing of Natural rubber latex into Constant viscosity and low viscosity rubbers.
- 4.1.4. Describe the processing of Natural rubber latex into OENR,DPNR and TPNR.
- 4.1.5. Define epoxidised NR, powdered NR, Cyclised NR and Chlorinated NR.
- 4.1.6. Explain the preparation and advantages of Tyre rubber, Latex master batch and powdered Rubber.
- 4.1.7. Define Reclaim rubber.
- 4.1.8. Describe the Reclamator and Digester process for the production of reclaimed rubber.
- 4.1.9. Discuss the advantages and applications of reclaimed rubbers.
- 4.1.10. State the types of Reclaimed rubber and its specifications.

CONTENT DETAILS

MODULE-I

History sources and important of natural rubber

Natural rubber- uniqueness, major sources statistics of production and consumption –India and World. Rubber yielding plants –propagation methods-different clones ,extraction of latex- tapping –standard of tapability -tapping task – tapping rest-tapping system .Types of tapping –Ladder ,slaughter and puncture tapping .Tapping systems – tapping efficiency –yield stimulants and rain guarding

MODULE-II

Preservation and concentration of NR latex.

Definition of NR latex –composition and colloid nature –role of micro organisms – pre coagulation -short term and long term preservation systems –Anti coagulants and preservatives Concentration of NR latex-creaming , centrifuging , -principle process machinery and plant lay out . BIS specification of centrifuged latex .Natural rubber latex –Double centrifuged latex , skim latex-skim rubber.

MODULE-III

Processing of NR latex in to dry marketable forms

Sheet rubber -Ribbed smoked sheets, Air dried sheets Processing, statistics ,Grades –Green book. Crepe rubber –processing , different grades .ISNR-Processing , machinery –grades –BIS specification.

MODULE-IV

Speciality forms of natural rubber and reclaim rubber Speciality rubbers -importance –features and applications processing –advantages and disadvantages -SP rubbers ,CV and LV rubbers ,OENR,DPNR, TPNR, Epoxidised NR, Powdered NR, Cyclised NR, Chlorinated NR, Tyre rubber , Latex master batch. Reclaim rubber – Reclamator process, Digester process –Advantages and applications .Types of reclaim rubbers –Specifications

REFERENCES

1. Hand book of natural rubber production –Rubber board
2. High polymer lattices vol 1 –D C Blackley (Publisher – Springer Netherlands, 1997 Edition)
3. Latex in industry. R.J Noble (Publisher – Rubber Age)